

AI-Driven Multimodal Stress and Emotion Analysis with Explainable Intelligence

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Project Proposal Report

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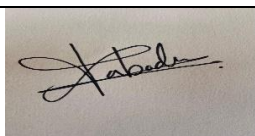
Sri Lanka Institute of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Stress has become one of the most widely recognized public health concerns of the contemporary era, affecting millions of individuals across occupational, academic, and personal domains worldwide. Traditional methods of stress assessment such as structured clinical interviews and standardized self-report questionnaires are inherently subjective, time-consuming, and largely impractical for continuous real-world monitoring. Recent advances in artificial intelligence and speech processing have opened promising new pathways for automated, non-invasive stress detection; however, most existing Speech Emotion Recognition (SER) systems stop classifying an emotional state without establishing a meaningful, psychologically grounded link to stress.

A fundamental conceptual gap persists in the current literature: the majority of SER systems treat emotion and stress as interchangeable constructs, which they demonstrably are not. Appraisal Theory is a well-established psychological framework that defines stress as the outcome of a person's cognitive evaluation of a situation relative to their perceived coping capacity, not merely as a function of the emotional state expressed. This psychologically important distinction has rarely been formalized within computational SER pipelines.

This proposal presents Component C2 with a broader four-component AI-driven multimodal stress and emotion analysis system. C2 proposes a pipeline that trains a SER model exclusively on the SAVEE (Surrey Audio-Visual Expressed Emotion) dataset, extracts log-mel spectrogram features combined with MFCCs and pitch/energy cues, and maps the classified emotion together with an arousal gate to a calibrated stress score grounded in Appraisal Theory. Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to mel-spectrograms to produce visual time-frequency heatmaps that explain which voiced regions most strongly influenced the model's prediction, thereby enhancing transparency and clinical trustworthiness.

The system is implemented using Python, librosa, torchaudio, and PyTorch, with training conducted on Google Colab. Expected outcomes include a well-validated SER model with competitive classification accuracy, a psychologically interpretable stress score, and explainable heatmaps suitable for use by mental health practitioners and digital well-being platforms. The outputs of C2 are integrated with other system components and provided to Component C4, which generates contextually appropriate supportive dialogue responses.

Keywords: Speech Emotion Recognition, Appraisal Theory, Stress Detection, Grad-CAM, Mel-Spectrogram, SAVEE Dataset, Explainable Artificial Intelligence

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AIMS	Autonomous Intelligent Machines and Systems
API	Application Programming Interface
CNN	Convolutional Neural Network
F1	F1 Score (Harmonic Mean of Precision and Recall)
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
IT	Information Technology
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficients
SAVEE	Surrey Audio-Visual Expressed Emotion
SDG	Sustainable Development Goal
SER	Speech Emotion Recognition
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

Stress is one of the most pervasive health challenges of the modern era. Globally, hundreds of millions of individuals are affected by stress-related disorders, and this burden has grown substantially in the aftermath of the COVID-19 pandemic, which accelerated rates of anxiety, burnout, and emotional dysregulation across multiple demographic groups [1]. In occupational settings, work-related stress accounts for a significant proportion of absenteeism, reduced productivity, and long-term mental illness, imposing enormous economic and social costs on both individuals and organizations [2]. In Sri Lanka, growing awareness of mental health issues particularly among university students and working professionals has highlighted the urgent need for accessible, low-cost, and non-invasive tools capable of detecting stress early and enabling timely intervention [3].

The voice is one of the most immediate and natural channels through which a person's internal emotional and physiological state is expressed. Changes in speaking rate, pitch, energy distribution, and vocal timbre have been consistently and reliably linked to variations in arousal and emotional state [4]. As a result, automated speech analysis has attracted considerable research attention as a practical pathway for detecting psychological stress without the requirement for wearable sensors, clinical laboratory conditions, or structured self-report assessments. Speech Emotion Recognition (SER) is the subfield of affective computing dedicated to inferring emotional states from the acoustic and linguistic properties of human speech [5].

Despite the substantial progress made in SER research over the past decade, a fundamental conceptual limitation persists across the majority of published systems: they conflate emotion with stress, treating the two as interchangeable constructs. Appraisal Theory as formalized in cognitive psychology by Lazarus and Folkman explicitly defines stress as the product of a person's cognitive appraisal of a situation as threatening or exceeding their perceived coping resources [6]. This distinction is not merely academic; it has direct practical consequences. A person may exhibit vocal markers of excitement or strong enthusiasm emotions characterized by high physiological arousal without experiencing any stress whatsoever. Equally, a person suffering from chronic, suppressed stress may speak with a largely neutral or subdued emotional tone. Conflating these two constructs leads to both false positives and false negatives in stress detection systems, reducing their clinical and applied utility.

Existing SER systems, both academic prototypes and commercially available products embedded in contact center analytics or wellness platforms, typically terminate at an emotion label output and do not produce a calibrated stress estimate [7]. Furthermore, most such systems offer no explanations for their predictions, operating as black boxes that cannot be interrogated, challenged, or audited. This opacity is a significant barrier to adoption in clinical and regulated environments, where transparency, accountability, and traceability of algorithmic decisions are essential requirements.

Component C2 of this project directly addresses these limitations by proposing an SER pipeline trained exclusively on the SAVEE (Surrey Audio-Visual Expressed Emotion) dataset [8]. The pipeline extracts log-mel spectrogram representations together with MFCCs and pitch/energy features, trains a CNN-LSTM or Transformer-based classifier to recognize the seven emotion categories present in SAVEE, and subsequently applies an Appraisal Theory-informed mapping module. This module combines the predicted emotion class with an arousal gate computed from pitch and energy features to generate a calibrated binary stress score. Gradient-weighted Class Activation Mapping (Grad-CAM) is then applied to the convolutional layers of the trained model, producing interpretable time-frequency heatmaps that visualize the spectral and temporal regions of the utterance that most significantly influenced the model's decision [9].

Component C2 is part of a broader four-component AI-driven multimodal stress and emotion analysis system developed by a collaborative research group. Component C1 handles physiological stress detection from wearable EEG and GSR sensor data. Component C3 performs text-based stressor identification and cognitive distortion detection using transformer-based NLP models. Component C4 is a dialogue system that forecasts the next emotional state of the interlocutor and generates contextually appropriate, strategy-guided supportive responses. The data flow within the system is structured such that the outputs of C1, C2, and C3 are integrated and provided to Component C4's emotion forecasting module. C3 also feeds its stressor and distortion labels into C4, completing the multimodal information pathway. The interaction between C2 and C4 is therefore architecturally critical: the stress score and emotion classification produced by C2 serve as primary inputs to the dialogue forecaster, enabling C4 to tailor its supportive responses to the user's current vocal-emotional state.

The societal relevance of this work extends well beyond academic interest. The United Nations Sustainable Development Goal 3: Good Health and Well-being explicitly call for the promotion of mental health and well-being across all populations [10]. Automated, accessible, and explainable stress detection tools represent one practical and scalable pathway through which technology can contribute to this goal, particularly in healthcare contexts where specialist resources are limited or geographically inaccessible.

1.2 Literature Review

The past five years have witnessed considerable and accelerating progress in both SER methodology and the application of explainability techniques to audio-based machine learning models. The following review situates Component C2 within the current state of the literature, drawing exclusively on the sources identified and provided in the project documentation.

Waleed et al. (2025) presented a hybrid SER pipeline evaluated on the MELD and RAVDESS datasets, combining convolutional feature extraction with sequential modeling architectures to capture the temporal dynamics inherent in continuous speech. Their work demonstrated clear value in multi-dataset evaluation protocols and highlighted the practical challenge of cross-corpus generalization, a limitation directly relevant to any system trained on a single corpus such as SAVEE, and one that motivates careful test set design [1].

Kim and colleagues (2025) proposed a SER system that applied Grad-CAM directly to time-domain speech representations, producing post-hoc visual explanations of the model's internal

decision process. Their findings confirmed that Grad-CAM is a viable, practical, and computationally efficient XAI technique for speech classification models, delivering meaningful interpretability without substantially increasing inference overhead [2].

Babko (2024) explored SER using pre-trained transfer learning architectures augmented with LIME and SHAP-based feature explanations. This work underscored the rapidly growing demand for interpretability in affective computing systems, observing that black-box emotion classifiers, however accurate, face meaningful resistance to adoption in both clinical settings and regulated enterprise environments [3].

Singh et al. (2023) introduced Constant-Q Transform (CQT) modulation spectral features for SER and reported performance improvements over standard MFCC-based approaches on several benchmark corpora. While their feature engineering contributions are technically noteworthy, the study did not attempt to address the downstream mapping of emotion classifications to stress estimates, leaving precisely the gap that Component C2 is designed to fill [4].

Meng et al. (2023) conducted a comprehensive and systematically organized survey of audio-domain explainability techniques, including gradient-based attribution methods such as Grad-CAM, integrated gradients, and attention-based visualizations. Their taxonomy provides a theoretically grounded basis for the selection of Grad-CAM as the explanatory technique for C2, noting that it requires no architectural modification to the base model and operates directly on the learned feature maps of the convolutional layers [5].

The PyTorch Grad-CAM framework, as documented and made publicly accessible by Gildenblat (2021), provides the practical reference implementation that Component C2 adapts for mel-spectrogram visualization. When applied to spectrograms, Grad-CAM highlights specific time-frequency bins corresponding to voiced segments with pitch or energy characteristics that most strongly influenced the emotion classification decision [6].

Kim et al. (2024), working in the domain of physiological stress detection using single-channel EEG and GSR signals in a virtual reality interview paradigm, demonstrated that deep learning models combined with post-hoc calibration and explainability layers can produce robust and trustworthy stress classifications. While their chosen modality differs from speech, their methodological approach, particularly their use of calibrated probabilistic outputs and multi-level SHAP explanations directly informs the design philosophy of C2 [7].

Inamdar et al. (2023) provided a comprehensive review of machine learning approaches to stress detection across wearable devices and social media text, synthesizing findings from multiple modalities and identifying common methodological weaknesses. Among the most frequently cited weaknesses were the use of small or lab-controlled datasets, an absence of real-world validation, and a near-universal lack of explanation mechanisms in published systems [8].

Ometov et al. (2025) reviewed the landscape of open-access stress and emotion datasets and noted the continued dominance of acted speech corpora including SAVEE, RAVDESS, and TESS in SER benchmarking research. Their analysis supports the use of SAVEE as the primary training corpus in C2 while acknowledging the inherent limitations of acted data for real-world

deployment, limitations that motivate the use of calibration techniques in the proposed pipeline [9].

Taken collectively, the reviewed literature confirms three important and consistent findings. First, CNN and LSTM-based architectures and increasingly Transformer variants represent the current state of the art for SER tasks on mel-spectrogram representations. Second, Grad-CAM is the most widely adopted and practically accessible technique for producing meaningful visual explanations in audio classification pipelines. Third, and most critically, no reviewed system explicitly formalizes the mapping from emotion classification to a psychologically grounded stress score using a framework such as Appraisal Theory, leaving a well-defined conceptual and technical gap that directly motivates the research design of Component C2.

1.3 Research Gap

The reviewed literature reveals three interconnected and compounding gaps that collectively define the research opportunity that Component C2 addresses.

The first and most fundamental gap is the absence of a psychologically grounded bridge between emotion classification and stress estimation. The SER literature has produced increasingly accurate emotion classifiers across multiple architectures and datasets; yet these systems consistently terminate at an emotion label. The clinical and cognitive psychology literature, by contrast, has long recognized that stress is a higher-order construct that arises from a person's appraisal of a situation as threatening in relation to their available coping resources. No existing SER system reviewed in this study formally operationalizes this distinction by incorporating an arousal gate and an appraisal-based mapping module to convert emotion probability outputs into a calibrated, psychologically meaningful stress score.

The second gap concerns the limited application of interpretable, post-hoc explanations within speech-based stress detection systems as opposed to general emotion recognition. While Grad-CAM has been extensively applied to image classification tasks and has begun to appear in general SER research, its application to mel-spectrograms in the specific and clinically relevant context of stress detection where the explanation must be tied not only to the emotion but to the derived stress score remains largely unexplored. Providing visual time-frequency heatmaps that explain which acoustic regions of a spoken utterance contributed to a stress estimate is a distinct contribution with direct practical relevance for clinical transparency and operator oversight.

The third gap relates to the structural integration of SER-derived stress signals into a broader conversational AI pipeline. Most SER systems operate as isolated modules, producing a label or score that is not consumed by or coordinated with any downstream application. The design of Component C2 as a structured, interface-defined module that feeds a calibrated, explainable stress signal into the dialogue forecasting and response generation pipeline of Component C4 represents an architectural contribution that addresses this integration gap and demonstrates the practical utility of the C2 outputs beyond standalone classification.

Component C2 addresses all three of these gaps through a combined technical and conceptual contribution: a SAVEE-trained SER model with Appraisal Theory-based stress mapping, Grad-

CAM-generated spectrogram heatmaps, calibrated confidence outputs, and a clearly defined output interface that connects seamlessly to Component C4 of the multimodal system.

2. OBJECTIVE

2.1 Main Objective

To develop, validate, and explain a Speech Emotion Recognition (SER) system trained on the SAVEE dataset that maps emotional classifications to psychologically grounded stress scores using Appraisal Theory. The system incorporates Grad-CAM visualizations on mel-spectrograms and integrates the calibrated stress output as a structured feed into Component C4 of the multimodal stress and emotion analysis system.

2.2 Specific Objectives

1. To preprocess the SAVEE dataset by resampling audio to 16 kHz mono, applying overlapping sliding window segmentation, and extracting log-mel spectrogram, MFCC, and pitch/energy features using the librosa and torchaudio libraries in the Google Colab environment.
2. To design and train a CNN-LSTM or Transformer-based SER classifier on the seven emotion categories of the SAVEE dataset, employing focal loss to address class imbalance and achieving a macro F1-score of at least 0.60 on the held-out test partition.
3. To implement an Appraisal Theory-informed emotion-to-stress mapping module that combines the predicted emotion class probability distribution with an arousal gate derived from pitch and energy features, producing a calibrated binary stress score (high or low) with an associated confidence probability.
4. To apply Grad-CAM to the convolutional layers of the trained SER model and generate interpretable heatmap overlays on mel-spectrograms, identifying and visualizing the specific time-frequency regions that most significantly influenced the stress prediction for each input utterance.
5. To evaluate the complete C2 pipeline using standard quantitative metrics including accuracy, per-class precision, recall, macro F1-score, ROC-AUC, and Expected Calibration Error and conduct a systematic ablation study to quantify the individual contribution of the arousal gate to stress prediction performance.
6. To define and implement a structured output interface for Component C2 that exposes the emotion class, calibrated stress score, and confidence probability as a standardized payload consumable by the Component C4 dialogue forecasting module within the four-component multimodal system.

3. METHODOLOGY

3.1 Research Approach

This research adopts an Experimental Research Methodology to systematically design, implement, and evaluate the proposed system. This approach is appropriate because the primary scientific contribution of Component C2 involves the comparison and rigorous validation of machine learning architecture and algorithmic techniques under controlled experimental conditions. The evaluation is conducted using clearly defined hypotheses, standardized performance metrics, and a fixed, publicly available dataset.

Specifically, the research tests the hypothesis that a CNN-LSTM or Transformer architecture trained on SAVEE log-mel spectrogram features, combined with an Appraisal Theory–informed arousal gate and stress mapping module, can produce a more psychologically meaningful and clinically interpretable stress estimate than a conventional emotion classifier that outputs only an emotion label. The hypothesis is examined through controlled experiments conducted on the SAVEE dataset, repeated training runs using different random seeds, and quantitative evaluation metrics including classification accuracy, F1-score, ROC-AUC, and Expected Calibration Error.

The experimental design follows an empirical validation process in which a publicly available dataset is used, multiple models are compared against a defined baseline, and the results are supported by reproducible quantitative evidence. The research contribution therefore extends beyond simple system development to include formalized model comparison, ablation analysis, and the integration of a psychologically grounded emotion-to-stress mapping module, thereby establishing the work within an empirical machine learning research framework.

3.2 Dataset

The SAVEE (Surrey Audio-Visual Expressed Emotion) dataset is used as the sole training and evaluation corpus for Component C2. SAVEE consists of acted emotional speech samples recorded from four male British English speakers across seven emotion categories: anger, disgust, fear, happiness, neutral, sadness, and surprise. The dataset contains a total of 480 utterances and is publicly accessible via Kaggle under an academic research license. Its manageable scale, clean labeling, and widespread use as a benchmark in the SER literature make it well suited for student-scale experimentation within the computational constraints of Google Colab.

The dataset will be partitioned into training (70%), validation (15%), and test (15%) subsets using a speaker-independent split to ensure that the model is evaluated on utterances from speakers it has not encountered during training, which provides a more realistic and rigorous evaluation protocol than a random split. Class imbalance across emotion categories, where the neutral class is overrepresented relative to classes such as disgust and fear, will be addressed using focal loss during model training.

3.3 Audio Preprocessing

All audio files are loaded and resampled to 16 kHz mono using the librosa library. Short-Time Fourier Transforms (STFT) will be applied using a window size of 25 ms and a hop length of 10 ms to produce time-frequency representations. Log-mel spectrograms will be extracted with 128 mel filter banks and converted to a logarithmic scale to produce log-mel representations that closely approximate the perceptual processing of the human auditory system. Mel-Frequency Cepstral Coefficients (40 coefficients) will be computed as supplementary representations. Pitch contours and frame-level energy values will be extracted using librosa's fundamental frequency estimator (YIN algorithm) and root mean square energy function. These pitch and energy values are retained separately from the spectrogram features and are used exclusively as inputs to the arousal gate computation in the stress mapping module.

3.4 Model Architecture

Two candidate architectures are implemented, trained, and compared. The primary architecture is a CNN-LSTM hybrid: stacked convolutional layers with batch normalization and ReLU activation functions extract spatial features from the mel-spectrogram across the frequency axis, while bidirectional LSTM layers subsequently model the temporal dynamics of the feature sequence along the time axis. The secondary architecture is a lightweight Transformer encoder, applied to the flattened and positionally encoded spectrogram sequence, providing an alternative mechanism for capturing long-range temporal dependencies. Both architectures will be constrained to fewer than one million parameters to ensure that training is feasible within the memory and compute limitations of the Google Colab free-tier GPU environment.

Focal loss will be employed in place of standard categorical cross-entropy loss to ensure that minority emotion classes which are underrepresented in SAVEE receive proportionally higher gradient signal during training, reducing the model's tendency to ignore infrequent classes. Dropout regularization (dropout probability $p = 0.3$) will be applied after each LSTM and fully connected layer to reduce overfitting on the relatively small dataset.

3.5 Appraisal Theory Stress Mapping

Following emotion classification, the predicted emotion probability distribution is passed through the Appraisal Theory-informed stress mapping module. This module operates through a transparent, three-stage logic. In the first stage, the dominant emotion class, with the highest predicted probability, is identified. In the second stage, an arousal level is computed from the pitch and energy values extracted during preprocessing: utterances with elevated average pitch and high root mean square energy are assigned a high-arousal designation, while utterances with low pitch and subdued energy are assigned a low-arousal designation. In the third stage, the combination of emotion class and arousal level is evaluated against a predefined appraisal mapping table derived from the psychological literature on Appraisal Theory, which assigns each emotion-arousal pair a binary stress indicator.

As a concrete illustration of the mapping logic: anger paired with high arousal is assigned a high-stress designation, reflecting the appraisal of a threatening stimulus combined with elevated physiological mobilization; calm or neutral speech paired with low arousal is assigned a low-stress designation; happiness with moderate arousal is treated as low or no stress; while

fear with high arousal is assigned high stress. This rule-based mapping is deliberately transparent, auditable, and easily modifiable by domain experts, ensuring that the stress scoring logic can be explained and challenged without requiring access to the internal model weights.

The final output of the mapping module is a calibrated binary stress label, high or low, accompanied by a confidence probability derived from the emotion classifier's SoftMax output. Post-training probability calibration will be performed using temperature scaling, a lightweight technique that adjusts the model's confidence output to better align with empirical accuracy on the validation set, improving the reliability of the confidence estimates for downstream consumption by Component C4.

3.6 Explainability (Grad-CAM on Mel-Spectrograms)

Gradient-weighted Class Activation Mapping (Grad-CAM) will be applied to the final convolutional layers of the trained CNN component. Grad-CAM operates by computing the gradient of the predicted class score with respect to the activations of the target convolutional layer and weighting those activation maps by the corresponding gradient magnitudes. The resulting weighted activation map is upsampled to the dimensions of the input mel-spectrogram and overlaid as a color heatmap, highlighting the time-frequency regions corresponding to specific voiced intervals and spectral frequency bands that most significantly influenced the model's classification decision.

The PyTorch-GradCAM library will be used to implement this pipeline. Heatmaps will be generated for each test utterance, enabling both qualitative inspection of individual predictions and systematic analysis of which acoustic features the model has learned to associate with each emotion category. This dual-level interpretability at the individual prediction level and at the class-level average provides a rich explanatory layer that accompanies every stress prediction made by the C2 pipeline.

3.7 Tools and Platforms

Tool / Platform	Purpose
Python	Primary programming language used for implementing preprocessing, model development, training, and evaluation code.
librosa	Used for audio loading, resampling, MFCC extraction, and pitch and energy computation during audio preprocessing.
torchaudio	Provides GPU-accelerated audio processing, spectrogram generation, and efficient audio data loading.
PyTorch	Deep learning framework used for defining model architecture, implementing training loops, and performing inference.
PyTorch-GradCAM	Library used to generate Grad-CAM visual explanations from convolutional feature maps of the trained model.

scikit-learn	Used for computing evaluation metrics, performing train/test dataset splitting, and implementing calibration utilities.
Google Colab	Cloud-based development environment providing GPU resources for training and experimentation (Pro).
Kaggle	Platform used for hosting and accessing the SAVEE dataset used in the research.

Table 1: Tools and Platforms

3.8 Ethical Considerations

The SAVEE dataset consists entirely of acted emotional speech produced by consenting adult male speakers in a controlled recording environment. It contains no sensitive personal data, no personally identifiable information beyond speaker identifiers used for evaluation splits, and no clinical patient records. All data is sourced from Kaggle under its published academic terms of use and will not be redistributed or used for any purpose beyond this research. No primary data collection from human participants is conducted as part of this research project; formal ethics board approval for participant recruitment is therefore not required.

The researcher acknowledges the broader ethical implications that arise when stress detection systems are deployed in practice, particularly with regard to informed consent, data privacy, the potential for algorithmic bias, and the risk of misuse in surveillance or employment screening contexts. These concerns inform the design choice to build transparent, explainable outputs through Grad-CAM heatmaps and an auditable appraisal mapping table, rather than deploying opaque black-box predictions. Responsible deployment of the technology developed in this research would require additional ethical review and governance frameworks beyond the scope of this academic project.

3.9 Validation Strategy and Performance Metrics

The following quantitative metrics will be computed on the held-out test partition of SAVEE to validate the C2 pipeline:

- Classification Accuracy: the proportion of test utterances whose emotion label is correctly predicted.
- ROC-AUC: the area under the receiver operating characteristic curve for the binary high/low stress classification task.
- Expected Calibration Error (ECE): a metric quantifying the alignment between the model's predicted confidence scores and empirical accuracy.
- Confusion Matrix: a full cross-tabulation of predicted versus true emotion labels used to identify systematic misclassification patterns.

A baseline comparison will be conducted using a classical MFCC feature vector combined with a Support Vector Machine (SVM) classifier. This baseline comparison provides quantitative evidence of the value added by the proposed deep learning architecture. A targeted ablation study will additionally assess the contribution of the arousal gate to stress prediction performance, comparing the accuracy and ROC-AUC of the full Appraisal Theory mapping

module against a version of the module that assigns stress based on emotion class alone, without the arousal component.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Four Component Multimodal System

The broader research project comprises four interdependent but independently developable components that together form a multimodal stress and emotion analysis system. Component C1 detects physiological stress from wearable EEG and GSR sensor data using a calibrated CNN-LSTM model with subject-level SHAP explainability. Component C2, the subject of this proposal, detects stress from speech using SER and Appraisal Theory-based mapping. Component C3 analyzes written text for stressor categories and cognitive distortions using a dual-head transformer model. Component C4 is the dialogue system that integrates inputs from C1, C2, and C3, forecasts the next emotional state of the user, selects an appropriate supportive strategy, and generates a contextually informed response using a fine-tuned language model.

The flow of data and control within the system follows a clearly defined architecture. Both C1 and C2 independently produce calibrated stress outputs that are fed into the emotion forecasting module of C4. C3 similarly feeds its stressor category and cognitive distortion labels into C4. C4 then integrates all three input streams to determine the appropriate dialogue strategy and generate the supportive response. Component C2 therefore occupies a critical position within the system architecture: its calibrated stress score and emotion probability vector are primary determinants of the emotional context available to C4 when selecting and generating a supportive reply.

4.2 Component C2 Internal Architecture

The internal processing pipeline of Component C2 consists of five sequential and clearly defined stages, each with a specific function and a well-defined input-output interface.

Stage 01 - Audio Collection and Preprocessing: Raw audio files from the SAVEE dataset are loaded using librosa, resampled to 16 kHz mono, and divided into overlapping windows. This stage normalizes the raw audio input into a consistent representation suitable for downstream feature extraction, regardless of the original recording conditions or file format.

Stage 02 - Feature Extraction: Log-mel spectrograms (128 mel filter banks), MFCCs (40 coefficients), and pitch/energy contours are extracted from each audio window. The log-mel spectrogram serves as the primary two-dimensional input feature map to the deep learning classifier, while pitch and energy values are retained as separate scalar features for use in the Stage 4 arousal gate computation.

Stage 03 - SER Model (CNN-LSTM or Transformer): The log-mel spectrogram tensor is passed through the trained CNN-LSTM or Transformer-based encoder, which produces a softmax probability distribution over the seven SAVEE emotion categories. Focal loss during training ensures that underrepresented emotion classes receive sufficient gradient signal, and dropout regularization reduces overfitting on the small dataset.

Stage 04 - Appraisal Theory Stress Mapping Module: The dominant predicted emotion class and the arousal gate value computed from the pitch and energy features extracted in Stage 2 are jointly evaluated against the predefined appraisal mapping table. This module outputs a binary stress label and an associated calibrated confidence score. Temperature scaling is

applied to the confidence probability to ensure that the model's expressed confidence is reliably aligned with its empirical accuracy.

Stage 05 - Grad-CAM Explainability Module: Grad-CAM is computed for the convolutional layers of the Stage 3 SER model, generating a heatmap overlaid on the input log-mel spectrogram. This visualization identifies the specific time-frequency regions most influential to the emotion classification and, by extension, to the stress prediction. The heatmap is stored as a structured output and made available for logging, clinical review, or operator inspection.

4.3 Component Interactions and Data Flow

The output of the Appraisal Theory Stress Mapping Module, comprising the calibrated binary stress label, the emotion probability vector, and the confidence score is formatted as a structured JSON payload and passed to the emotion forecasting module of Component C4. C4's Transformer Encoder-based forecaster ingests this payload alongside the current dialogue history to predict the probable next emotional state of the user and select a contextually appropriate supportive response strategy. The Grad-CAM heatmap generated in Stage 5 is stored as a visual log associated with each prediction event, enabling post-hoc auditing and transparent review of the system's reasoning by authorized operators or clinical supervisors.

4.4 Scalability and Deployment Considerations

The CNN-LSTM model is constrained to fewer than one million parameters and is exported to ONNX (Open Neural Network Exchange) format upon completion of training. ONNX export enables lightweight inference on CPU environments without requiring PyTorch or GPU hardware, making the component deployable in resource-constrained settings such as mobile applications or edge devices. The sliding-window preprocessing pipeline is designed to process audio in near real-time, with a target inference latency of under 150 ms per window on CPU, satisfying the non-functional performance requirements defined in Section 5. The modular, interface-defined design of C2 ensures that the component can be upgraded, retrained, or replaced independently without disrupting the broader four-component multimodal pipeline.

SPEECH EMOTION RECOGNITION STRESS DETECTION SYSTEM

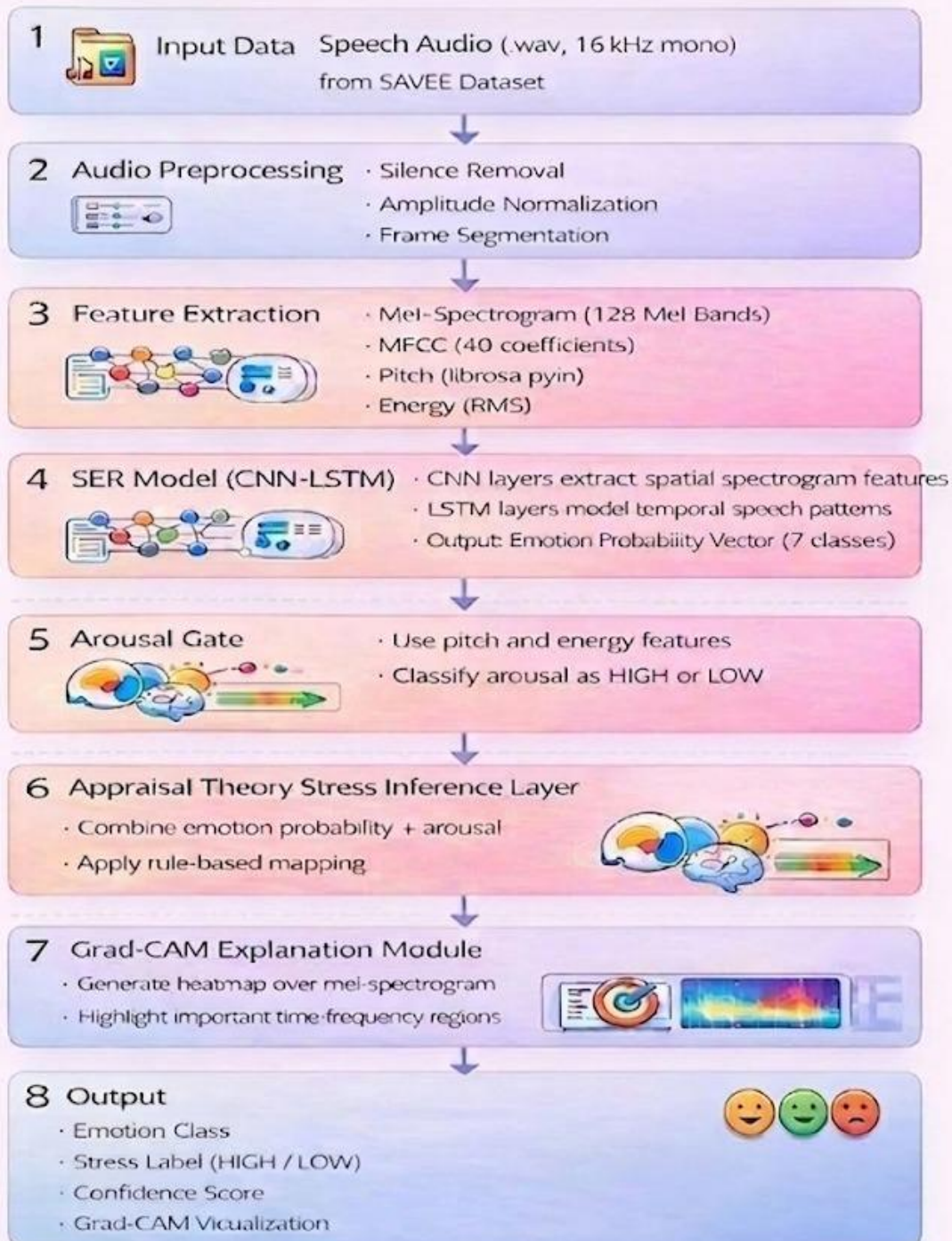


Figure 1: High-Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The following stakeholders interact with or benefit from the C2 speech-based stress detection module.

Stakeholder	Role
Users / Speakers	Provide speech audio input and receive detected emotion and stress analysis results from the system.
Mental Health Professionals	Use speech-based stress indicators to understand emotional and stress patterns.
System Administrators	Manage system deployment, monitoring, and maintenance of the stress detection module.
Researchers / Developers	Train models, evaluate system performance, and improve speech-emotion recognition algorithms.

Table 2: Stakeholder identification

5.2 Functional Requirements

FR	Functional Requirement
FR1	The system shall use speech audio samples from the SAVEE dataset as the primary dataset for model training and evaluation.
FR2	The system shall preprocess audio signals by re-sampling to 16 kHz, normalizing amplitude, and removing noise.
FR3	The system shall extract acoustic features including mel-spectrograms, MFCC, pitch, and energy from the audio input.
FR4	The system shall classify speech emotions using deep learning models such as CNN-LSTM or Transformer architectures.
FR5	The system shall estimate an arousal score using pitch and energy features derived from the audio signal.

FR6	The system shall map emotion probabilities and arousal levels to a stress score using Appraisal Theory.
FR7	The system shall classify the predicted stress level into high-stress or low-stress categories.
FR8	The system shall generate Grad-CAM visual heatmaps on mel-spectrograms to explain model predictions.
FR9	The system shall output the emotion label, calibrated stress score, prediction confidence and Grad-CAM heatmap for visualization and for integration with Component C4.
FR10	The system shall allow researchers to evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.

Table 3: Functional Requirements

5.3 Non-Functional Requirements

NFR	Category	Requirement
NFR1	Performance	The system should process audio input and produce emotion and stress predictions within a few seconds per audio input.
NFR2	Scalability	The architecture should allow scaling to larger datasets or cloud deployment (e.g., AWS or GCP) if required.
NFR3	Security	Audio data should be processed securely and handled according to ethical data usage guidelines.
NFR4	Usability	The system should provide a simple interface (e.g., Google Colab notebook or web interface) for uploading audio and viewing results.

NFR5	Reliability	The system should consistently produce stable predictions when evaluated on repeated experiments.
NFR6	Maintainability	The system should follow a modular architecture so that components such as feature extraction, model architecture, or explainability modules can be updated independently.

Table 4: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the technology developed in Component C2 encompasses digital mental health and well-being platform developers, corporate employee wellness solution providers, and healthcare technology companies seeking to integrate non-invasive speech-based stress monitoring into their product offerings. A secondary target market includes academic research institutions requiring validated, explainable SER components for affective computing research pipelines, and telehealth service providers seeking to augment remote consultations with automated, interpretable emotional state monitoring capabilities.

A representative customer persona is a mid-sized employee wellness software company that wishes to differentiate its platform by adding a voice-based stress detection feature to its mobile application. Such a company seeks to provide its users, primarily working professionals, with real-time feedback on their stress levels based on brief spoken interactions, enabling timely self-management and escalation to human support when warranted.

6.2 Value Proposition

Component C2 offers a technically differentiated value proposition built on three interconnected pillars. The first pillar is psychological grounding: the system produces a stress estimate, not merely an emotion label, derived from a formally specified Appraisal Theory mapping, making its output directly relevant and actionable for well-being applications where stress is the target construct. The second pillar is transparency: Grad-CAM heatmaps make the system's reasoning process visible, auditable, and defensible, a critical requirement for clinical adoption and regulated enterprise deployment. The third pillar is operational accessibility: the lightweight model design and ONNX export capability make the system feasible to run on mobile and edge devices without dependency on cloud-hosted GPU infrastructure.

6.3 Revenue Model

The commercialization pathway envisions a Software-as-a-Service API licensing model in which licensees pay either a per-query fee or a monthly subscription to access the C2 inference endpoint. For each submitted audio input, the API endpoint would return the predicted emotion class, calibrated stress score, confidence probability, and an optional Grad-CAM visualization. An enterprise licensing tier would provide access to the model weights and the contractual right to fine-tune the model on proprietary datasets. Academic and non-commercial research institutions would be offered access under a separate non-commercial research license with a significantly reduced or waived fee.

6.4 Competitive Advantage

The commercially available SER systems embedded in contact center analytics and enterprise communications platforms typically provide only a raw emotion classification or a simple valence score, without any stress estimation, psychological grounding, or explanatory output. Component C2 differentiates itself on all three of these dimensions: the Appraisal Theory-based stress mapping produces outputs that are clinically meaningful rather than merely

technically accurate; the Grad-CAM heatmaps provide a layer of transparency that competing systems do not offer; and the modular integration with a multimodal pipeline that incorporates physiological, speech, and text signals positions the technology within a more comprehensive and holistic well-being monitoring ecosystem than any single-modality competitor.

6.5 Intellectual Property Considerations

The primary intellectual property assets created by this research are the Appraisal Theory-informed emotion-to-stress mapping module, the integrated CNN-LSTM SER and Grad-CAM explanation pipeline, the calibration methodology applied to stress confidence scores, and the structured output interface connecting C2 to C4. These contributions may be protected through academic publication, which establishes a documented and dated record of prior art, or through a provisional patent application if the research is subsequently taken forward to commercial product development. The SAVEE dataset is used under its published academic research license and does not constitute proprietary intellectual property of the research team.

7. BUDGET AND JUSTIFICATION

Item	Description	Estimated Cost (USD)	Justification
Google Colab Pro Subscription	Monthly subscription for GPU acceleration (approx. USD 10/month, two semesters $\approx$ 10 months)	100	Enables accelerated training and experimentation without purchasing hardware.
Storage & Data	Kaggle dataset storage	0	The SAVEE dataset is hosted on Kaggle and is freely downloadable.
Software	Python, Librosa, Torchaudio, PyTorch, Scikit-learn	0	Open-source licenses allow free use in research.
Internet Cost	Internet connectivity required for downloading datasets, conducting experiments, and accessing cloud-based resources.	50	Covers connectivity for research and data downloads.
	Total	150	

Table 5: Budget Estimation

8. WORK BREAKDOWN STRUCTURE (WBS)

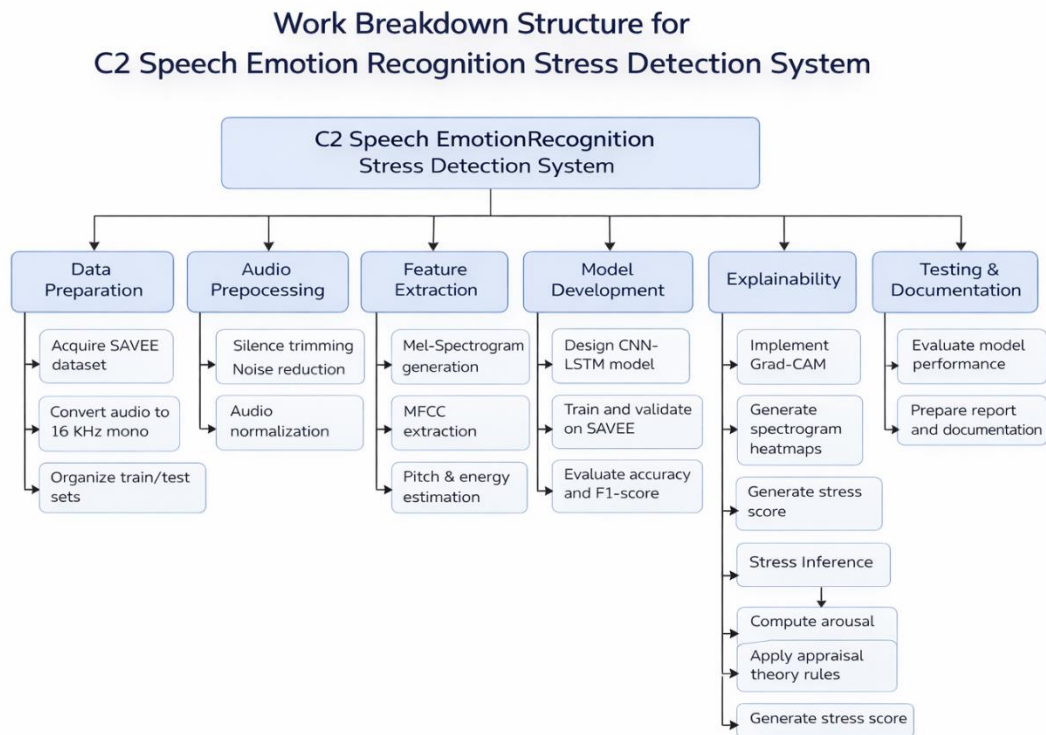


Figure 2: Work Breakdown Structure

9. GANTT CHART

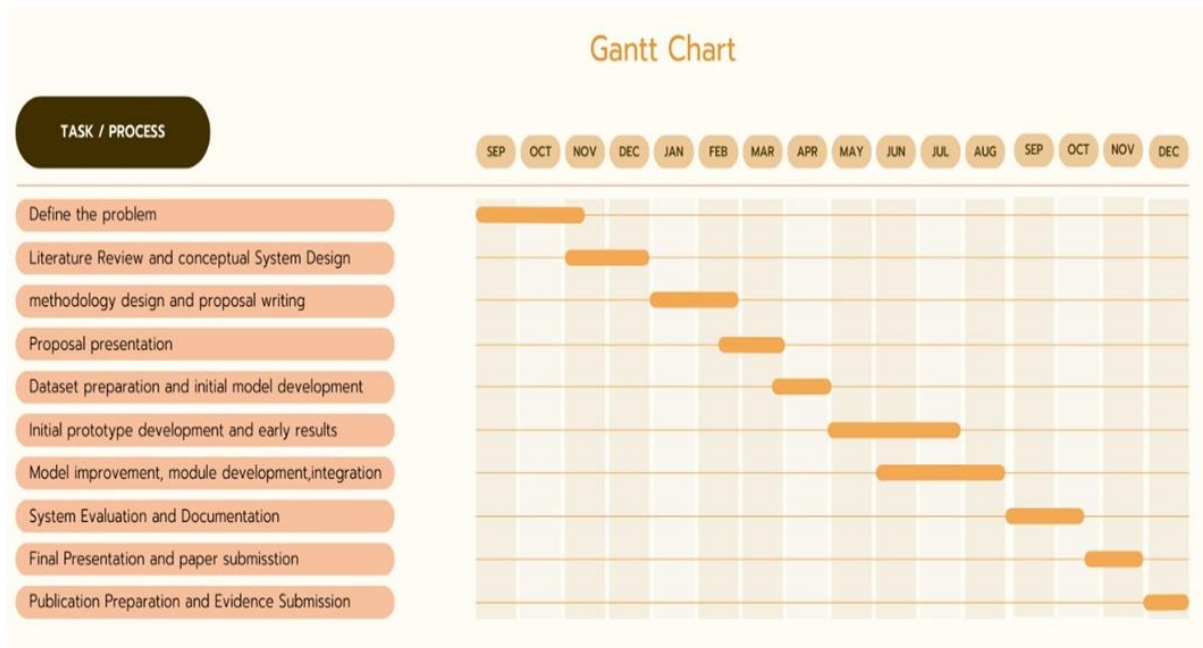


Figure 3: Gantt Chart

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11. LIST OF APPENDICES

11.1 Appendix A – Risk Analysis

Risk Description	Likelihood	Impact	Mitigation Strategy
Model overfitting on the small SAVEE dataset (480 utterances)	High	High	Apply dropout ($p=0.3$), Spec Augment time/frequency masking, pitch-shift data augmentation, and early stopping with validation F1 monitoring.
Google Colab session timeout during long training runs	Medium	Medium	Save model checkpoints to Google Drive every 5 epoch; resume training from last checkpoint.
Class imbalance causing poor F1 on minority emotion classes	Medium	High	Implement focal loss and optional oversampling of minority classes (disgust, fear, surprise).
Grad-CAM heatmaps providing visually uninformative or diffuse activations	Low	Medium	Evaluate Grad-CAM at multiple convolutional layers and select the layer with the highest spatial specificity; use Grad CAM++ as a fallback.
Appraisal mapping table producing stress labels inconsistent with clinical expectations	Medium	High	Validate mapping tables against at least three published Appraisal Theory sources before deployment; document all mapping decisions.

Appendix A: Risk Analysis table